Free Model Stack — Model Guide

22 keyless models across the free frontier lane, the Kilo pool and failover combos — ranked for practical use in OpenCode

Generated 2026-11-05 · provider **00-free** (127.0.0.1:18900) · roster is live, re-run `npm run sync` after it changes · ids and numbers match the `/models` picker

1 How to read this

Every model below is served by the local **Our Free Model** gateway with **no login and no API key**. The rank is the picker order: a practical blend of **capability ceiling, context/output size and measured latency** — it is guidance for picking a model fast, not a benchmark result. Free lanes rotate: when one model returns 429/503, the combo or a router covers the same request.

> **Lane meaning.** `free` = OpenCode Zen free lane (frontier models, large injected system prompt ≈ 5.6–6.4k tokens per request). `kilo` = Kilo AI's keyless *isFree* pool (≈2k tokens overhead; **prompts may be logged upstream**). `combo` = a local failover chain, not a model. **Vision** = accepts images. **Out** = max output tokens — the column that quietly matters most for big rewrites and document generation.

2 Ranked roster (22 models)

#	MODEL	LANE	CONTEXT	OUT	VISION	SPECIALIZES IN
00	**Auto Combo** combo:recommended-free	combo	1M	131k	✓	Safest default: falls over to the next free link if one dies before answering. Use when a specific model keeps 429ing.
01	**DeepSeek V4.1 Flash** deepseek-v4.1-flash-free	free	1M	384k	✓	All-round flagship. Strong reasoning and agentic coding, enormous output budget — the default daily driver.
02	**GLM 5.3 Flash** glm-5.3-flash-free	free	1M	131k	✓	Fast general reasoning, strong multilingual work, vision included — great second opinion to 01.
03	**MiMo V2.6 Flash** mimo-v2.6-flash-free	free	1M	131k	✓	Balanced multimodal generalist; smooth on mixed text + image conversations.
04	**Nemotron 3.5 Super VL** nemotron-3.5-super-vl-free	free	1M	131k	✓	Vision-language specialist: screenshots, UI review, charts, dense OCR-style reading.
05	**Nemotron 3.5 Lightning** nemotron-3.5-lightning-free	free	128k	32k	—	Latency-first reasoning; snappy chat and tight tool loops where 128k context is enough.
06	**Kilo Auto** kilo-auto/free	kilo	256k	32k	—	Kilo's own task router across its free pool — good hands-off option, cheapest overhead on the Kilo side.
07	**Nemotron 3 Ultra 550B** nvidia/nemotron-3-ultra-550b-a55b:free	kilo	1M	65k	—	Biggest brain in the stack: hardest reasoning, deep analysis, long research tasks.
08	**Nemotron 3.5 Super VL (Kilo)** nvidia/nemotron-3.5-super-vl:free	kilo	1M	131k	✓	Vision workhorse with 1M context: diagram-to-code, screenshot debugging, chart extraction.

#	MODEL	LANE	CONTEXT	OUT	VISION	SPECIALIZES IN
09	**Nemotron 3.5 Lightning (Kilo)** nvidia/nemotron-3.5-lightning:free	kilo	1M	65k	—	Speed + 1M context: fast triage of very long logs, docs and diffs.
10	**Dots3 Note Preview** dots-studio/dots-3-note-preview:free	kilo	512k	460k	✓	Document-scale tasks: whole repo/book in, book-length output. Rare 460k output ceiling.
11	**Glyph Cluster** stealth/glyph-cluster	kilo	256k	256k	—	Massive transforms: full rewrites, exhaustive summaries, big translation passes.
12	**North Mini Code** cohere/north-mini-code:free	kilo	256k	64k	—	Compact coding specialist: refactors, code review, tight diff generation.
13	**OpenRouter Free Router** openrouter/free	kilo	200k	32k	✓	Routes across OpenRouter's own `:free` catalog — variety when Kilo's pool is busy.
14	**Big Pickle** big-pickle	free	262k	131k	✓	General chat with vision and a roomy output budget; comfortable everyday fallback.
15	**Beam** beam-free	free	131k	32k	—	General reasoning, mid-size; dependable mid-tier when the flagships queue up.
16	**Beam 1.0** reflection/beam-free	kilo	1M	131k	—	Reasoning at 1M context — long analysis without the free-lane overhead.
17	**Flint 40B** flint-40b-free	free	131k	32k	—	40B open reasoning, solid mid-tier judgment on everyday engineering questions.
18	**Pulse Alpha** pulse-alpha-free	free	131k	32k	—	Experimental generalist — a taste of upcoming free frontier models; expect volatility.
19	**Nemotron 3 Nano Omni** nvidia/nemotron-3-nano-omni-30b-a3b-reasoning:free	kilo	256k	65k	✓	Small omni reasoning + vision; thrifty for image-aware batch jobs.
20	**Crouton Alpha** crouton-alpha-free	free	131k	32k	—	Brand-new free-lane alpha — preview model; treat as experimental until probed.
21	**Crouton Alpha (Kilo)** stealth/crouton-alpha	kilo	1M	943k	✓	New Kilo-side alpha/preview entry — fresh free capacity; verify with a live prompt.

3 Field notes — what each model is for

00 · Auto Combo combo
Chains free links and only fails over when a link dies before producing content. For unattended jobs and a "just work" default.

01 · DeepSeek V4.1 Flash free
All-round reasoning and coding with the biggest output budget here (384k). The main driver for real coding sessions and long writes.

02 · GLM 5.3 Flash free
Fast, well-structured, strong multilingual coverage, vision. A second opinion, translations, quick screenshot reviews.

03 · MiMo V2.6 Flash free
Balanced multimodal generalist. UI iteration, screenshot → notes, everyday image + text chat.

04 · Nemotron 3.5 Super VL free
Reads images carefully — charts, tables, UI, OCR-style extraction. Diagram → spec, screenshot → bug report.

05 · Nemotron 3.5 Lightning free
The quickest free-lane feel on short context. Quick chat, small edits, tight tool loops (128k cap).

06 · Kilo Auto kilo
Routes inside Kilo's free pool with the leanest Kilo overhead. Hands-off default; prompts may be logged upstream.

07 · Nemotron 3 Ultra 550B kilo
The heaviest reasoning in the stack. Architecture calls, postmortems, hard debugging — slow, so not for churn.

08 · Nemotron 3.5 Super VL (Kilo) kilo
Same careful vision work without the Zen-lane prompt overhead. Image-heavy tasks when the free lane throttles.

09 · Nemotron 3.5 Lightning (Kilo) kilo
1M context at speed. First-pass triage of huge logs or diffs, then hand the finding to a deeper model (65k out).

10 · Dots3 Note Preview kilo
Document scale in *and* out — a 460k output ceiling. Whole books, reports, specs; long inputs restructured.

11 · Glyph Cluster kilo
Mass transformation: rewrite, translate, summarize large inputs with room to write back. Not a deep reasoner.

12 · North Mini Code kilo
Code review, refactors, tests, diffs. Purpose-built; pair with 01 when the task is half prose.

13 · OpenRouter Free Router kilo
Breadth across OpenRouter's own `:free` catalog. Variety and fallback when Kilo's pool is saturated.

14 · Big Pickle free
Roomy general chat with vision. Everyday Q&A, doc drafting, a comfortable fallback when flagships throttle.

15 · Beam free
Dependable mid-tier reasoning. A safe, consistent everyday choice rather than a specialist.

16 · Beam 1.0 kilo
Reasoning at 1M context with leaner overhead. Long analysis when the free lane is busy.

17 · Flint 40B free
40B open-model judgment. A second perspective on plans and mid-size engineering questions.

18 · Pulse Alpha free
A preview of upcoming free frontier models. Experimentation only — expect volatility.

19 · Nemotron 3 Nano Omni kilo
Small omni reasoning plus vision. Alt text, screenshot triage at volume, image-aware batch jobs.

20 · Crouton Alpha free
New free-lane alpha that appeared in the roster. Treat as a preview model: try it on a real task before relying on it.

21 · Crouton Alpha (Kilo) kilo
New Kilo-side alpha entry. Fresh free capacity on the Kilo lane; run a real prompt to see what it is good at.

4 Pick by task

IF YOU NEED…	USE	WHY
A default that just works	01 DeepSeek V4.1 Flash, or 00 Auto Combo	Best balance of reasoning, speed and 384k output
The hardest thinking	07 Nemotron 3 Ultra 550B, or 01 DeepSeek V4.1 Flash	Largest model in the stack, 1M context
Screenshots, UI, charts, OCR	04 Nemotron 3.5 Super VL, or 08 Nemotron 3.5 Super VL (Kilo)	Vision-language specialists with 1M context
Whole repo / book in one shot	10 Dots3 Note Preview, or 09 Nemotron 3.5 Lightning (Kilo), 11 Glyph Cluster	512k–1M input with up to 460k output
Coding & refactors	12 North Mini Code, or 01 DeepSeek V4.1 Flash	Purpose-built code models + long-output generalists
Agentic tool loops	06 Kilo Auto, or 01 DeepSeek V4.1 Flash	1M context, vision, 262k outputs
Bulk extraction / classification	19 Nemotron 3 Nano Omni, or 12 North Mini Code	Cheap, fast, purpose-shaped
Variety when Kilo is busy	13 OpenRouter Free Router, or 00 Auto Combo	Routers over independent pools

5 Measured latency & cost profile

LANE	OBSERVED	PROMPT OVERHEAD
00 combo	1.0–2.5 s	~5.9k tok
01 DeepSeek V4.1 Flash	1.2–1.4 s	~5.9k tok
06 Kilo Auto	1.1 s (direct lane)	~2k tok
fcc / OpenRouter `:free`	2.2–5.1 s	~22 tok

Reading the numbers. Free-lane models are fastest but carry a large injected system prompt (that's the cost of keyless frontier access); Kilo lanes are leaner; FCC via OpenRouter is the leanest and slowest. A one-line question costs ~5.9k tokens on the free lane — for cost-sensitive bulk work prefer the utility-tier kilo models or the `fcc` provider.

Ranking caveat. These are latency and size facts plus practical fit. No benchmark scores are implied; upstream rosters change and quotas rotate. Re-run `npm run bakeoff` any time for a fresh scorecard.

6 Using them in OpenCode

- **Select:** `/models` → provider **00-free** → `NN-name`. The numbers are the ranking.
- **Refresh the roster** after upstream changes: `npm run sync` (re-pulls live models, keeps this order).
- **Reorder the top**: edit the `PRIORITY` array in `bin/sync-models.mjs`, then `npm run sync`. Unlisted new models append alphabetically — nothing disappears.
- **Privacy:** Kilo-lane prompts may be logged upstream; route anything sensitive through your own aggregate keys (dashboard → Aggregate) or local Ollama via the `fcc` provider.
- **More lanes** (dashboard at `http://127.0.0.1:18900`): EAC (GitHub sign-in + star) and account channels (LobsterAI, Qoder, TRAE, Cline, Loomy, Raccoon, MiniMax Code, ZCode, Gemini) add models after you sign in.

7 Where this came from — and how it actually works

No model is hosted by this stack. Every answer comes from a real upstream you can audit. The two things that are local are a loopback gateway and the small repo that wires it into OpenCode.

LANE	UPSTREAM	HOW IT'S REACHED	GOOD TO KNOW
free	OpenCode Zen inference opencode.ai/inference/v1/*	`Authorization: Bearer public`, a minted per-request session id, and the official v2 client fingerprint	Direct upstream access — **not** a bridge through OpenCode. The fingerprint is why a one-line prompt costs ~5.9k tokens.
kilo	Kilo AI public gateway api.kilo.ai	Keyless; the live `isFree` slice of Kilo's model listing	Lean overhead (~2k tok). **Prompts may be logged upstream** — don't send secrets.
combo	This machine	Named failover chains defined in the gateway	No upstream of its own — it just tries the next free link when one dies before answering.
fcc	OpenRouter + your local Ollama	free-claude-code on 127.0.0.1:8082 (Anthropic Messages)	Leanest prompts (~22 tok). Optional second gateway, not required for anything above.

> **The local pieces. Our Free Model** (Ebony-Vinyl/dsh-our-free-model, MIT) runs as a loopback service on `127.0.0.1:18900` and speaks OpenAI Chat, OpenAI Responses and Anthropic Messages. **free-claude-code** (Alishahryar1, AGPL-3.0) runs on `127.0.0.1:8082`, installed at runtime via uv — never bundled here. **free-model-bridge** (KoBadger, MIT) ships **no gateway code at all**: just setup, sync, verify, bake-off and this guide. Upstream is cloned straight from GitHub, so what runs on your machine is auditable. No affiliation with OpenCode, DeepSeek, Ebony-Vinyl or Alishahryar1. The bare `dsh-our-free-model` package on npm is a third-party repackage — this repo clones the source instead.

> **Security posture.** Loopback only, LAN relay disabled, gateway API key kept in `~/.our-free-model/settings.json` as `forwardKey`, and `HOST=127.0.0.1` pinned in `~/.fcc/.env` (the FCC default is `0.0.0.0` — LAN-exposed with no auth). Nothing needs admin rights; autostart uses your Startup folder, not Task Scheduler.

8 Get it running from the repo

Prereqs: Node ≥ 22.19 (or ≥ 24) and git. Optional: Bun for a faster free-lane transport, and uv (it fetches Python 3.14 itself) only if you want FCC.

```
git clone https://github.com/KoBadger/free-model-bridge
cd free-model-bridge
npm run setup            # add -- --with-fcc for the second gateway
```

STEP	WHAT `NPM RUN SETUP` DOES
1 · fetch	Shallow-clones the OFM gateway into `~/Apps/our-free-model` (or `~/our-free-model` on macOS/Linux).

STEP	WHAT `npm run setup` DOES
2 · start	Starts the gateway on `127.0.0.1:18900` if it isn't up, and waits for `/health`.
3 · config	Creates `~/.config/opencode/opencode.json` if missing; a timestamped backup is written before *every* change.
4 · provider	Writes the `00-free` provider with the live roster and the `provider.use` allow entry. The digits-first id is why this group sits on top.
5 · autostart	Creates a Startup-folder shortcut (no admin). macOS/Linux prints a systemd user-unit line instead.
6 · optional FCC	Installs free-claude-code via uv on Python 3.14.10, pins `HOST=127.0.0.1`, starts it, wires the `fcc` provider and its shortcut.

Everyday flow.

- `/models` → `00-free` → `NN-name`.
- `npm run sync` — re-pull the live roster into OpenCode.
- `npm run guide` — regenerate this PDF and the field notes from that roster.
- `npm run bakeoff` — fresh latency + correctness scorecard.
- `npm run verify` — the full health check.

OpenCode hot-reloads config, so nothing needs a restart.

Unlock more lanes in the dashboard at `http://127.0.0.1:18900` (login token in `service.out.log`):

- **EAC** — GitHub sign-in + star (server-verified), co-paid frontier models.
- **Account channels** — LobsterAI, Qoder, TRAE, Cline, Loomy, Raccoon, MiniMax Code, ZCode, Gemini, Kilo.
- **Aggregate** — 76 providers with your own API keys, stored in a `0600` settings file.

After unlocking: `npm run sync` then `npm run guide`, so the picker and this PDF both show the new models.
